



Fundamentals of Optics

Lecturer

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Rules

- Final exam: 30%
- Mid-term exam: 30%
- Homework: 25%
- Attendance: 15%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

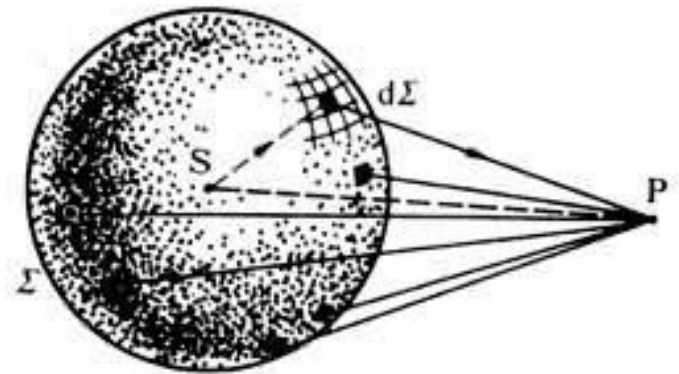
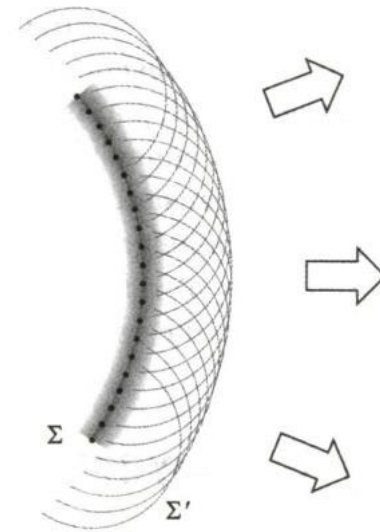
CHAPTER 9

DIFFRACTION

9.1 The Huygens-Fresnel Principle

Huygens's principle: every point on a propagating wavefront serves as the source of spherical secondary wavelets, such that the wavefront at some later time is the envelope of these wavelets.

Huygens-Fresnel principle: Every unobstructed point of a wavefront, at a given instant, serves as a source of spherical secondary wavelets (with the same frequency as that of the primary wave). The amplitude of the optical field at any point beyond is the superposition of all these wavelets (considering their amplitudes and relative phases)

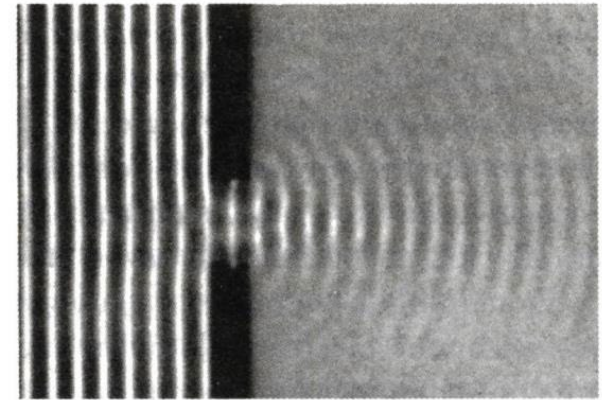
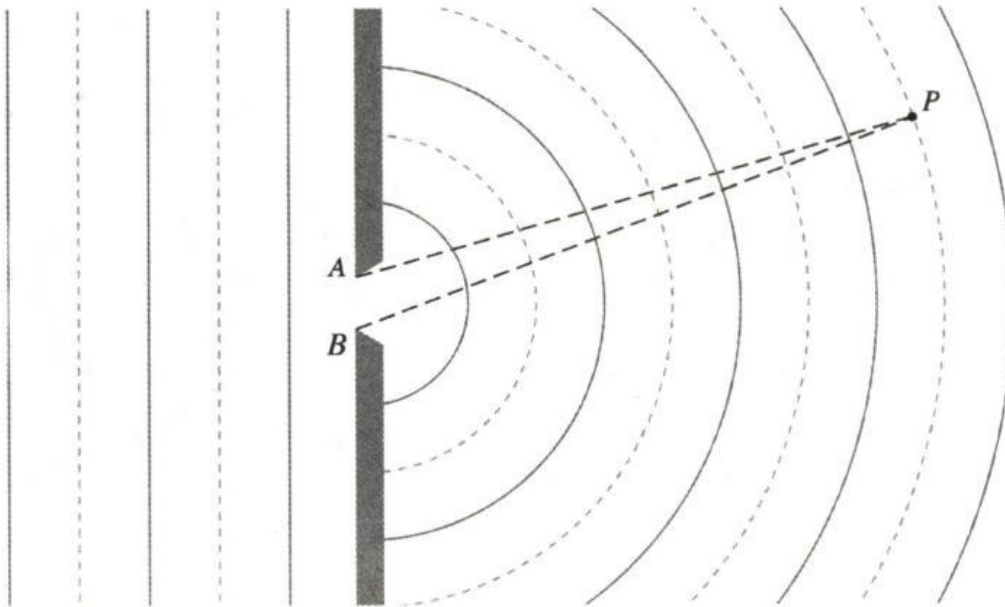


Example: ripple tank photograph:

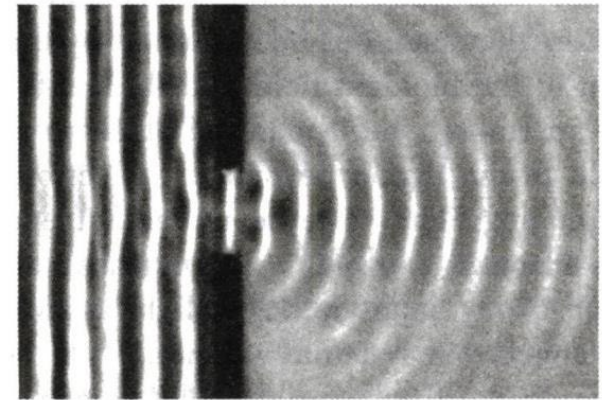
The maximum optical path length difference:

$$\Lambda_{max} = |AP - BP|$$

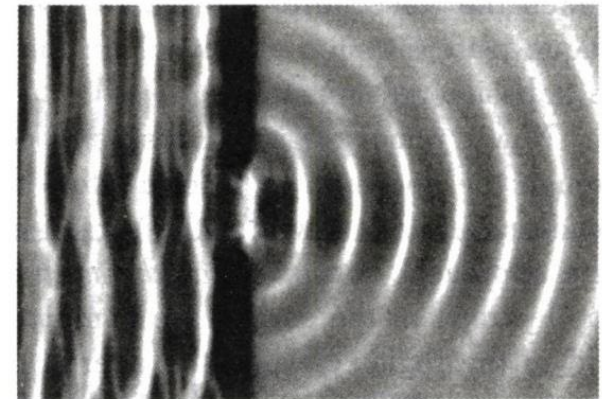
- I. $\lambda > AB$, namely $\lambda > \Lambda_{max}$: all interference is constructive whenever P happens to be.
- II. $\lambda < AB$: all the wavelets will interfere constructively in a small region where $\lambda > \Lambda_{max}$. Beyond this zone, some of wavelets can interfere destructively.



(a)

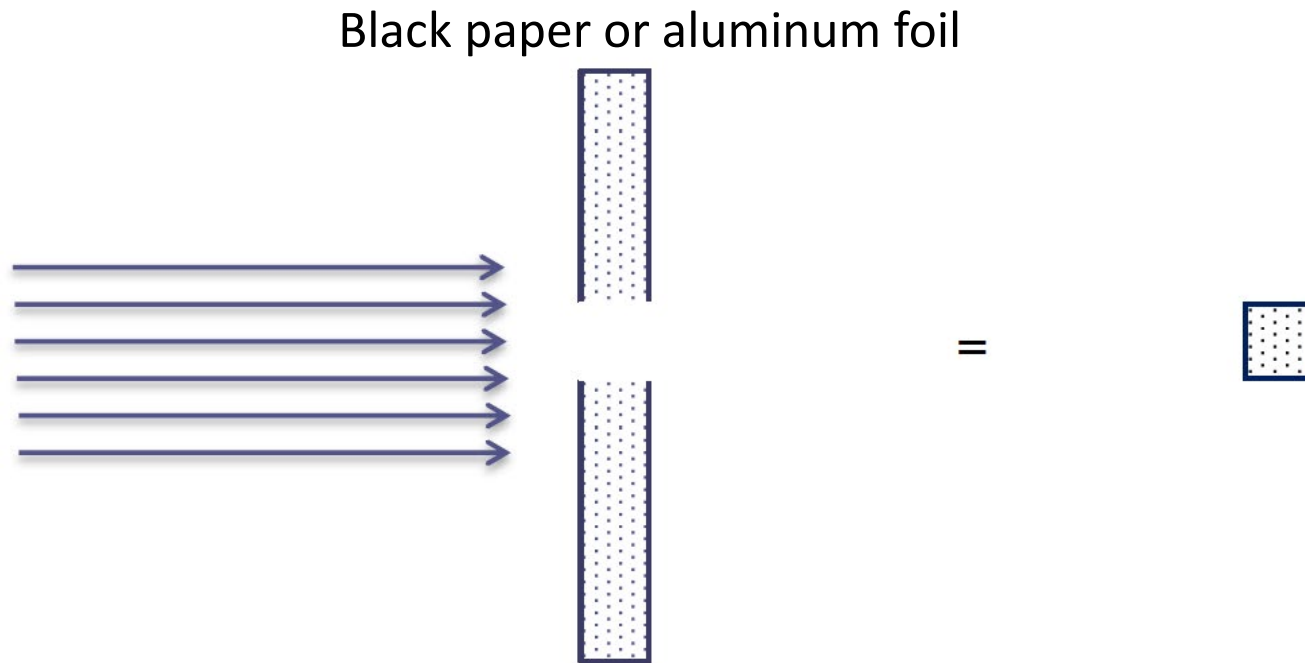


(b)



(c)

9.1.1 Opaque Obstructions



The diffraction field can be pictured as arising exclusively from a set of fictitious noninteracting oscillators distributed uniformly over the region of the aperture --- **Huygens-Fresnel Principle**.

9.1.2 Fraunhofer and Fresnel Diffraction

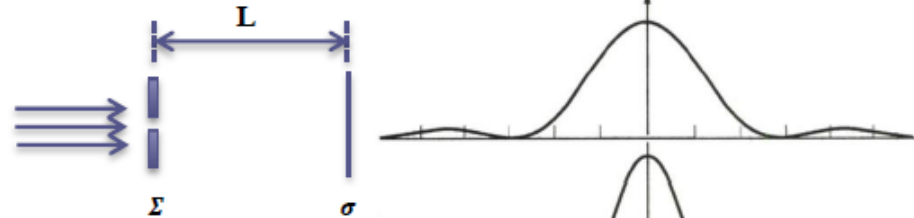
Fraunhofer (far-field) diffraction: at a long distance, the projected pattern have spread out considerably, bearing little or no resemblance to the actual aperture.

As long as both the incoming and outgoing waves approach being planar over the extent of the diffraction apertures, **Fraunhofer diffraction** obtains.

Fresnel (near-field) diffraction: the image of the aperture, though still easily recognizable, becomes increasingly more structured as the fringes become more prominent.

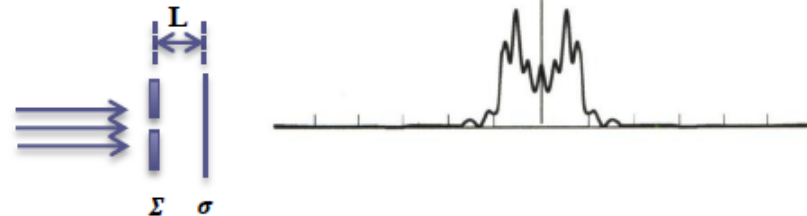
When the position of observation plane (σ) is fixed and a point source is moved toward (Σ), spherical waves would impinge on the aperture, and the **Fresnel pattern** would exist.

Fraunhofer (far-field) diffraction



$L \nearrow$

Fresnel (near-field) diffraction



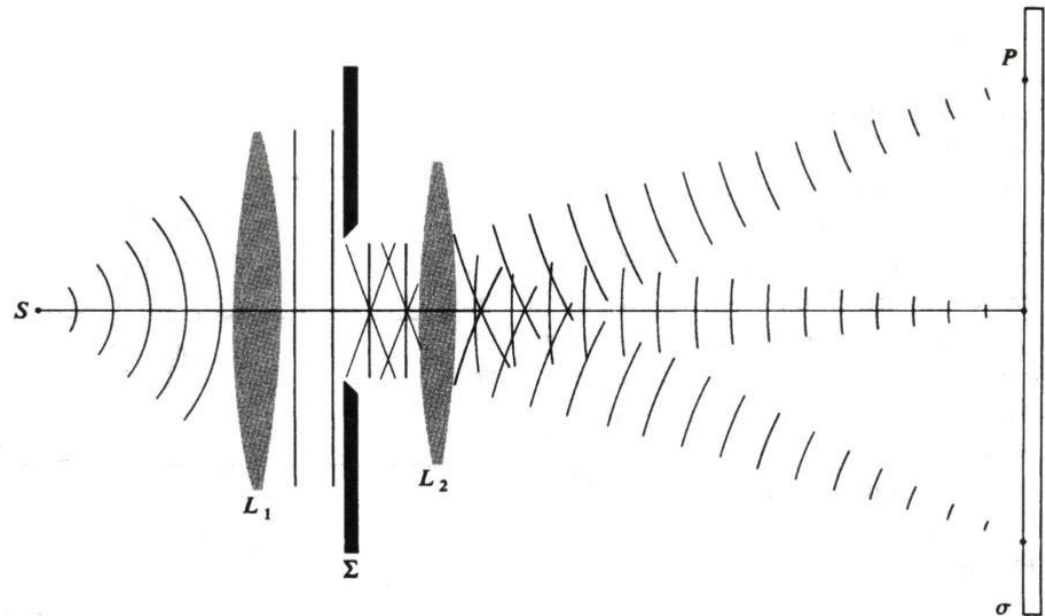
Fraunhofer diffraction: the wavefronts impinging on, and emerging from the aperture are planar. And the observation is at infinity. So the path difference could be described by a linear function of the two aperture variables.

Rule of thumb

$$R > a^2 / \lambda$$

a : the width of aperture.

R : the smaller one of the two distance from S to Σ and Σ to P .



Achieve Fraunhofer diffraction using lenses

Fresnel diffraction: a spherical wavefront illuminates the hole, and the distance from S to each point on the screen is different and field strength is also different. Much the same thing is true for the diffracted waves going from the aperture to P .

9.1.3 Several Coherent Oscillators

Condition: a linear array of N coherent point oscillator with no intrinsic phase difference.

The separate wave amplitudes arriving at P is equal:

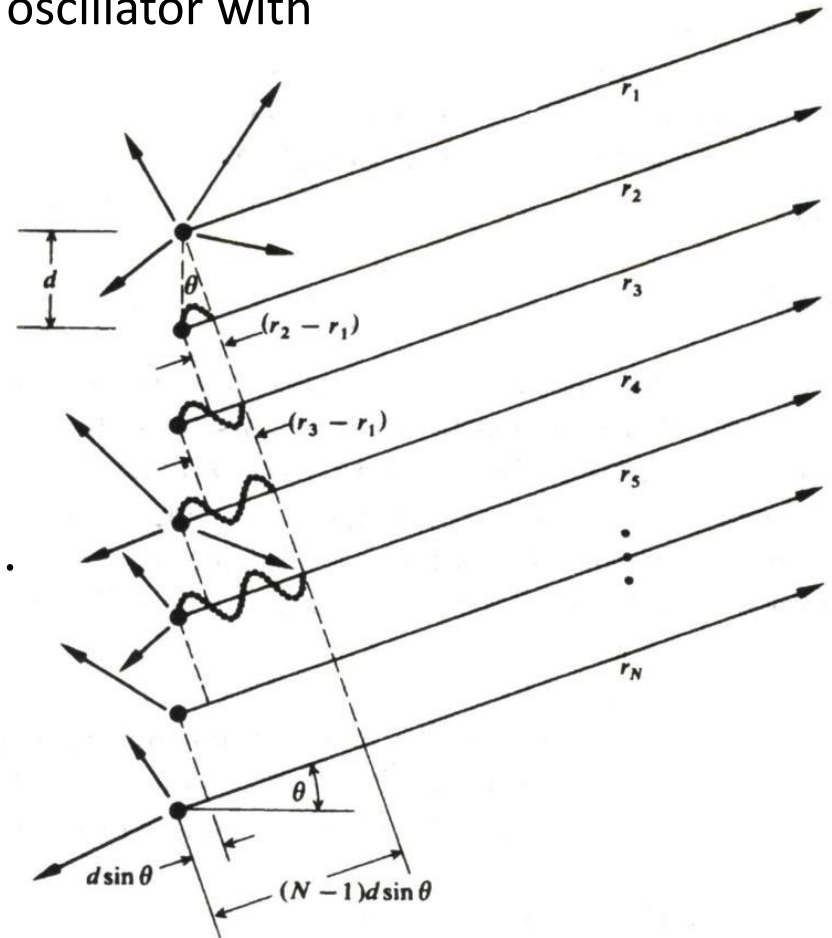
$$E_0(r_1) = E_0(r_2) = \dots = E_0(r_N) = E_0(r)$$

The sum of the interfering spherical wavelets at P

$$E = E_0(r)e^{i(kr_1 - \omega t)} + E_0(r)e^{i(kr_2 - \omega t)} + \dots + E_0(r)e^{i(kr_N - \omega t)}$$

$$E = E_0(r)e^{i(kR - \omega t)} \left(\frac{\sin N\delta/2}{\sin \delta/2} \right)$$

$$I = I_0 \left(\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} \right)$$



$$E = E_0(r)e^{i(kr_1-\omega t)} + E_0(r)e^{i(kr_2-\omega t)} + \dots + E_0(r)e^{i(kr_N-\omega t)}$$

$$E = E_0(r)e^{-i\omega t}e^{ikr_1}[1 + e^{ik(r_2-r_1)} + e^{ik(r_3-r_1)} + \dots + e^{ik(r_N-r_1)}]$$

The phase difference between two adjacent point sources is $\delta = k_0\Lambda$

Considering the refractive index (n) of the medium $\Lambda = nd\sin\theta$

$$\delta = k_0\Lambda = kdsin\theta$$

$$E = E_0(r)e^{-i\omega t}e^{ikr_1}[1 + (e^{i\delta}) + (e^{i\delta})^2 + (e^{i\delta})^3 + \dots + (e^{i\delta})^{N-1}]$$

$$1 + (e^{i\delta}) + (e^{i\delta})^2 + (e^{i\delta})^3 + \dots + (e^{i\delta})^{N-1} = \frac{e^{i\delta N} - 1}{e^{i\delta} - 1} = \frac{e^{i\delta N/2}[e^{i\delta N/2} - e^{-i\delta N/2}]}{e^{i\delta/2}[e^{i\delta/2} - e^{-i\delta/2}]}$$

$$E = E_0(r)e^{-i\omega t}e^{i[kr_1+(N-1)\delta/2]} \left(\frac{\sin N\delta/2}{\sin \delta/2} \right) = e^{i\delta(N-1)/2} \frac{\sin N\delta/2}{\sin \delta/2}$$

Define R as the distance from the center of the line of oscillators to the point-P

$$R = \frac{1}{2}(N-1)d\sin\theta + r_1$$

$$E = E_0(r)e^{i(kR-\omega t)} \left(\frac{\sin N\delta/2}{\sin \delta/2} \right) \quad I = I_0 \left(\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} \right)$$

The flux-density distribution:

$$I = I_0 \left(\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} \right) = I_0 \frac{\sin^2 \left[N \left(\frac{kd}{2} \right) \sin \theta \right]}{\sin^2 \left[\left(\frac{kd}{2} \right) \sin \theta \right]}$$

I_0 : flux density from any single source arriving at P

(I) The principal maxima: $N^2 I_0$.

$$\sin N\delta/2 \neq 0 \quad \text{and} \quad \sin \delta/2 = 0$$

$$\delta = 2m\pi, m = 0, \pm 1, \pm 2, \dots$$

(II) The minimum: zero

$$\sin N\delta/2 = 0 \quad \text{but} \quad \sin \delta/2 \neq 0$$

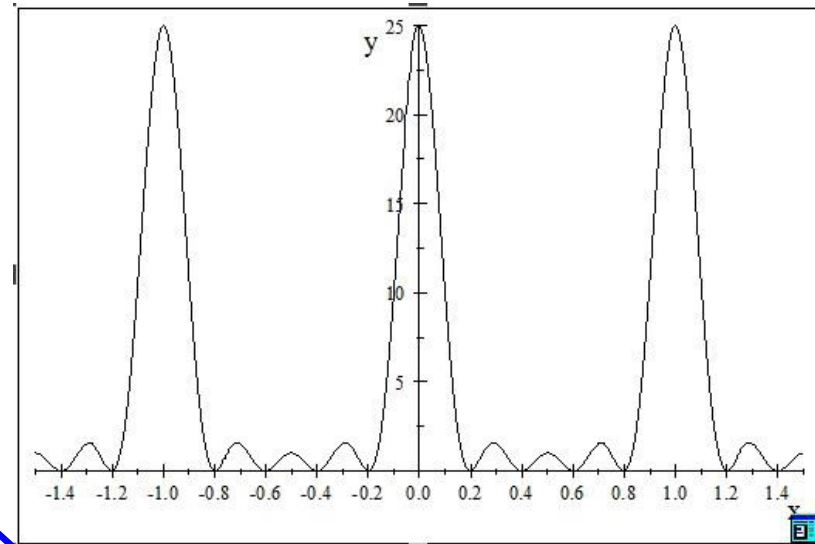
$$\frac{\delta}{2} = \pm \frac{\pi}{N}, \pm \frac{2\pi}{N}, \pm \frac{3\pi}{N}, \dots, \pm \frac{(N-1)\pi}{N}, \pm \frac{(N+1)\pi}{N}$$

(III) The (N-2) subsidiary maxima

$$\sin N\delta/2 = \pm 1$$

$$\frac{\delta}{2} = \pm \frac{\pi}{2N}, \pm \frac{3\pi}{2N}, \pm \frac{5\pi}{2N}, \dots$$

The expression gives rise to a series of sharp principal peaks separated by small subsidiary maxima.



L'Hospital's Rule
(洛必达法则)

When $\delta = 2m\pi$

$$\frac{\sin^2 N\delta/2}{\sin^2 \delta/2} = N^2$$

The idealized line source of electron-oscillator: the sources are very weak, their number (N) is tremendously large, and separation between them is vanishingly small.

Each point emits a spherical wavelet:

$$E = \left(\frac{\epsilon_0}{r} \right) \sin(\omega t - kr)$$

ϵ_0 : the source strength (波源强度).

Source number in a minute but finite segment Δy_i (the oscillators within Δy_i have a negligible relative phase difference):

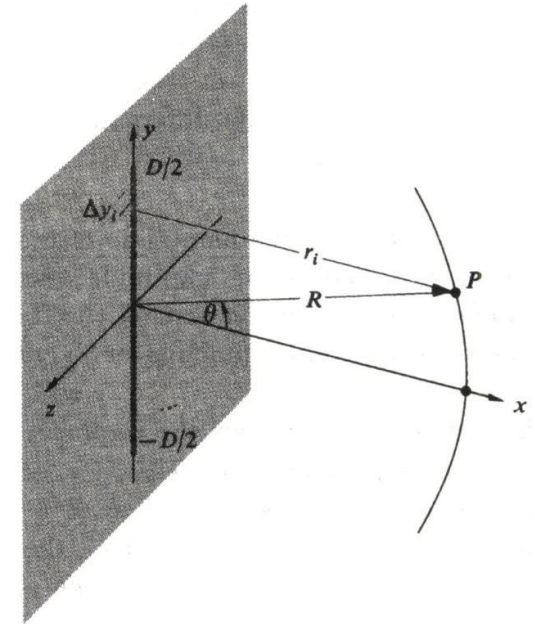
$$\frac{N\Delta y_i}{D} \quad \begin{array}{l} N \text{ is the total number of the electron-oscillators;} \\ D \text{ is the entire length of the line source.} \end{array}$$

Total M such segments, the contribution to electric-field intensity at P from the i -th segment:

$$E_i = \left(\frac{\epsilon_0}{r_i} \right) \sin(\omega t - kr_i) \left(\frac{N\Delta y_i}{D} \right)$$

Source strength per unit length for infinite N .

$$\epsilon_L \equiv \frac{1}{D} \lim_{N \rightarrow \infty} (\epsilon_0 N)$$



The net field at P :

$$E_i = \sum_{i=1}^M \left(\frac{\epsilon_L}{r_i} \right) \sin(\omega t - kr_i) \Delta y_i$$

For a continuous line source:

$$E_i = \epsilon_L \int_{-D/2}^{D/2} \frac{\sin(\omega t - kr_i)}{r} dy$$

Condition: the point of observation is very distant from the coherent line source and $R \gg D$: $r(y) \approx R$

$$dE = \frac{\epsilon_L}{R} \sin(\omega t - kr) dy$$

Expand $r(y)$:

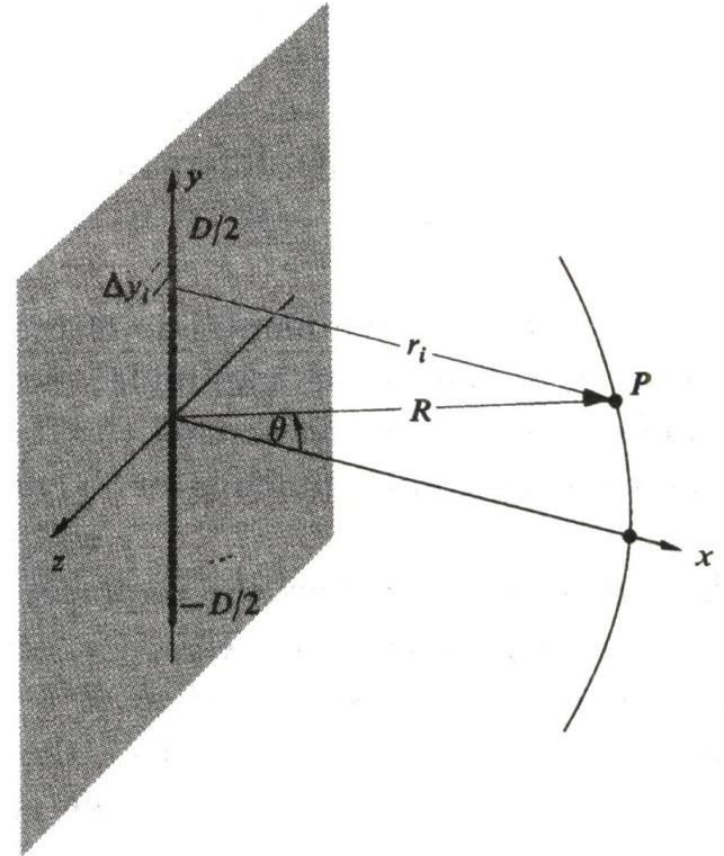
$$r = R - y \sin \theta + \left(\frac{y^2}{2R} \right) \cos^2 \theta + \dots$$

Fraunhofer condition: R is large enough

$$r = R - y \sin \theta$$

$$E = \frac{\epsilon_L}{R} \int_{-D/2}^{D/2} \sin[\omega t - k(R - y \sin \theta)] dy$$

$$E = \frac{\epsilon_L D}{R} \left(\frac{\sin \beta}{\beta} \right) \sin(\omega t - kR) \quad \beta \equiv \left(\frac{kD}{2} \right) \sin \theta$$



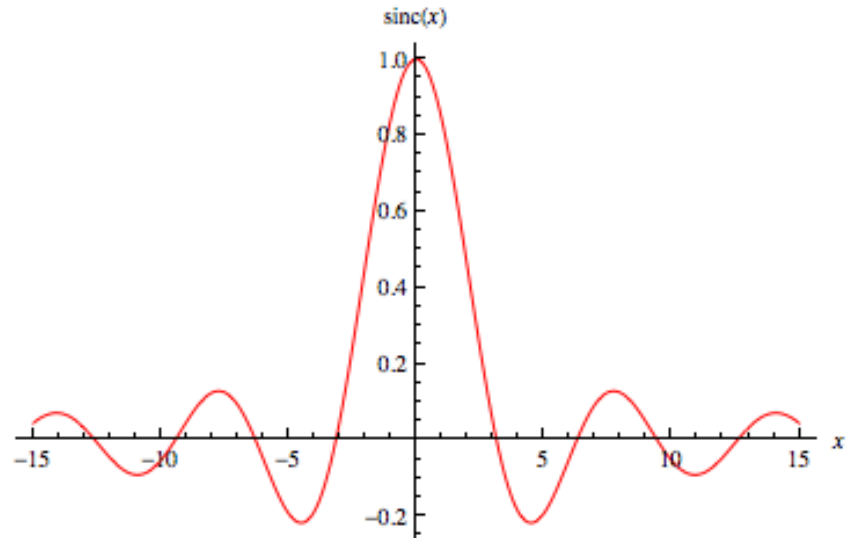
Irradiance: $I(\theta) = \langle E^2 \rangle_T$

$$I(\theta) = \left(\frac{\epsilon_L D}{R} \right)^2 \left(\frac{\sin \beta}{\beta} \right)^2 \langle \sin^2(\omega t - kR) \rangle_T = \frac{1}{2} \left(\frac{\epsilon_L D}{R} \right)^2 \left(\frac{\sin \beta}{\beta} \right)^2$$

$\theta = 0, \frac{\sin \beta}{\beta} = 1$ and $I(\theta) = I(0)$: the principal maximum

$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2$$

$$\beta \equiv \left(\frac{kD}{2} \right) \sin \theta = \left(\frac{\pi D}{\lambda} \right) \sin \theta$$



(I) $D \gg \lambda$:

The irradiance drops extremely rapidly as θ deviates from zero. So the relatively long coherent line source can be envisioned as a single-point emitter radiating in the forward direction ($\theta=0$).

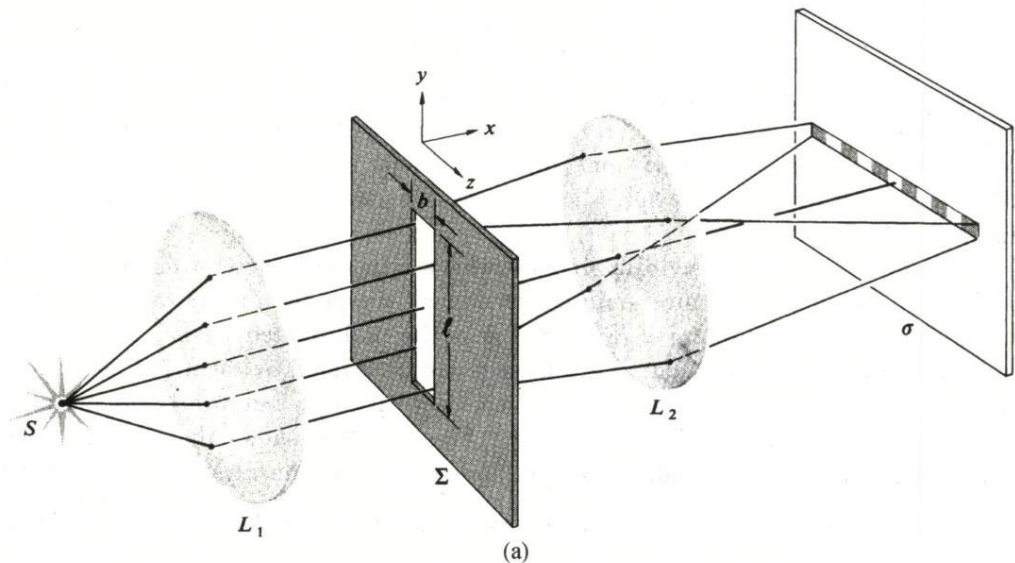
(II) $D \ll \lambda$: $\sin \beta \approx \beta, I(\theta) = I(0)$

The line source resembles a point source emitting spherical waves.

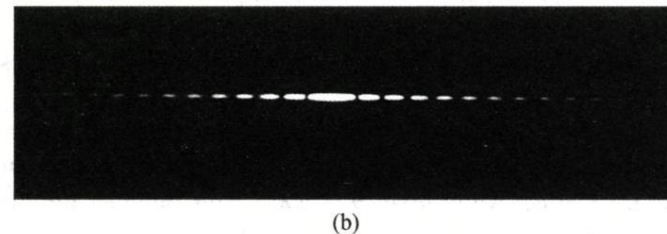
9.2 Fraunhofer Diffraction

The single slit Fraunhofer diffraction by a slit or elongated, narrow rectangular hole.

- ❑ Width: several hundred λ .
- ❑ Length: a few centimeters.



Each strip is a long coherent line source and can be replaced by a point emitter on the z -axis, which radiates a circular wave in the xz -plane.



The problem is reduced to that of finding the field in the xz -plane, once again, this is equivalent to a coherent line source.

$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta$$

The extrema of $I(\theta)$ occur at $dI/d\beta=0$

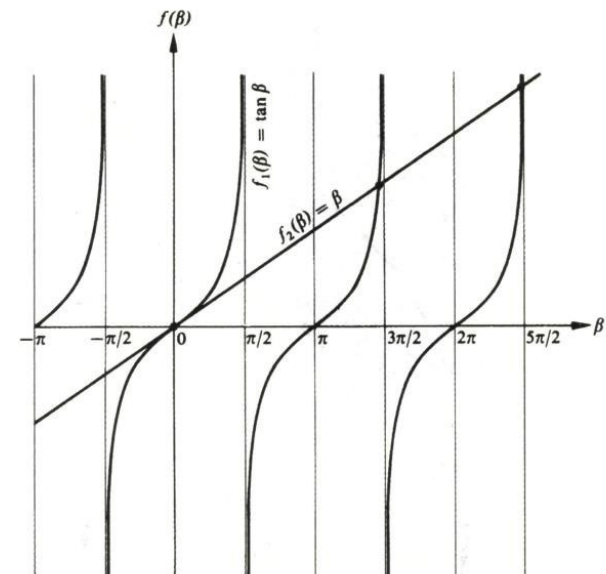
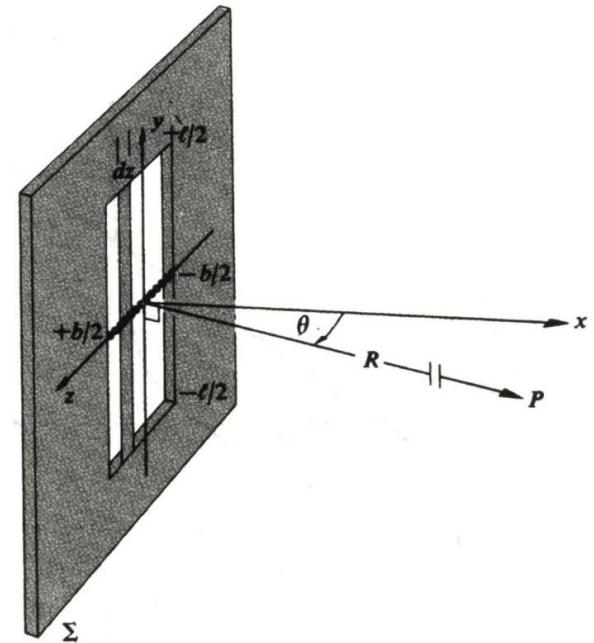
$$\frac{dI}{d\beta} = I(0) \frac{2\sin \beta (\beta \cos \beta - \sin \beta)}{\beta^3} = 0$$

□ The minima of irradiance: zero ($\sin \beta=0$):

$$\beta = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

□ The subsidiary maxima: $\beta \cos \beta - \sin \beta = 0 \Rightarrow \tan \beta = \beta$

$$\beta = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \dots$$



$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2$$

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